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DESIGN AND ANALYSIS OF ALGORITHM

competitive programming

QUESTION 1:

Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:
Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer:

```
#include <stdio.h>


int main() {
    int n;


    scanf("%d", &n);
```

```
    int a[n];


    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }


    // Compare every element with the elements after it
    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (a[i] == a[j]) {
                printf("%d", a[i]);
                return 0;
            }
        }
    }


    return 0;
}
```

RESULT:
Thus the code is verified and the expected output is obtained.

QUESTION 2:

Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

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First Line - Number of elements

n Lines - n Elements

Output Format:
Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer:

```
#include <stdio.h>


int main() {
    int n;
    scanf("%d", &n);


    int a[n];


    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);


    int slow = a[0];
    int fast = a[0];


    do {
        slow = a[slow];
        fast = a[a[fast]];
    } while (slow != fast);
```

```
    slow = a[0];


    while (slow != fast) {
        slow = a[slow];
        fast = a[fast];
    }


    printf("%d", slow);


    return 0;
}
```

RESULT:
Thus the code is verified and the expected output is obtained.

QUESTION 3:

Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Complexity

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

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The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer:

```
#include <stdio.h>


int main() {
    int T;
    scanf("%d", &T);
```

```
    while (T--) {
        int m, n;
        scanf("%d", &m);

        int a[m];
        for (int i = 0; i < m; i++)
            scanf("%d", &a[i]);

        scanf("%d", &n);

        int b[n];
        for (int i = 0; i < n; i++)
            scanf("%d", &b[i]);

        for (int i = 0; i < m; i++) {
            for (int j = 0; j < n; j++) {
                if (a[i] == b[j]) {
                    printf("%d ", a[i]);
                    break;
                }
            }
        }

        printf("\n");
    }

    return 0;
}
```

RESULT:

Thus the code is verified and the expected output is obtained.

QUESTION 4:

Print Intersection of 2 sorted arrays-O(m+n)Time Complexity,O(1) Space Complexity

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. **Line 1 contains N1, followed by N1 integers of the first array**
2. **Line 2 contains N2, followed by N2 integers of the second array**

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer:

```
#include <stdio.h>

int main() {

    int T;

    scanf("%d", &T);


    while (T--) {

        int m, n;

        scanf("%d", &m);


        int a[m];

        for (int i = 0; i < m; i++)

            scanf("%d", &a[i]);


        scanf("%d", &n);


        int b[n];
```

```
    for (int i = 0; i < n; i++)
        scanf("%d", &b[i]);

    int i = 0, j = 0;

    while (i < m && j < n) {
        if (a[i] == b[j]) {
            printf("%d ", a[i]);
            i++;
            j++;
        }
        else if (a[i] < b[j]) {
            i++;
        }
        else {
            j++;
        }
    }

    printf("\n");
  }

  return 0;
}
```

RESULT:
Thus the code is verified and the expected output is obtained.

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QUESTION 5:

Pair with Difference-O(n^2)Time Complexity,O(1) Space Complexity

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:
1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer:

```
#include <stdio.h>


int main() {
    int n, k;
```

```
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &k);

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i != j && a[j] - a[i] == k) {
                printf("1");
                return 0;
            }
        }
    }

    printf("0");

    return 0;
}
```

RESULT:
Thus the code is verified and the expected output is obtained.

QUESTION 6:

Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

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Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:
1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer:

```
#include <stdio.h>


int main() {
    int n, k;


    scanf("%d", &n);
```

```
    int a[n];


    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);


    scanf("%d", &k);


    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i != j && a[j] - a[i] == k) {
                printf("1");
                return 0;
            }
        }
    }


    printf("0");


    return 0;
}
```

RESULT:
Thus the code is verified and the expected output is obtained.